

Foundation (Jalapeno)

Q1. Which of the following functions has an inverse?

- (A) $f(x) = x^2$
- (B) $f(x) = x^3$
- (C) $f(x) = |x|$
- (D) $f(x) = 2x + 1$
- (E) $f(x) = x - 2$

Q2. Find the inverse of $f(x) = 3x - 2$. What is $f^{-1}(x)$?

- (A) $f^{-1}(x) = \frac{x+2}{3}$
- (B) $f^{-1}(x) = \frac{x-2}{3}$
- (C) $f^{-1}(x) = \frac{x+2}{2}$
- (D) $f^{-1}(x) = \frac{x-2}{2}$
- (E) $f^{-1}(x) = \frac{x+2}{4}$

Q3. Given $f(x) = 2x + 1$ and $g(x) = \frac{x-1}{2}$, which of the following is true?

- (A) $f(g(x)) = x$ and $g(f(x)) = x$
- (B) $f(g(x)) = x$ but $g(f(x)) \neq x$
- (C) $g(f(x)) = x$ but $f(g(x)) \neq x$
- (D) $f(g(x)) \neq x$ and $g(f(x)) \neq x$
- (E) None of the above

Q4. A company's profit P (in thousands of dollars) is given by $P(x) = 2x - 5$, where x is the number of products sold (in thousands). What is the inverse function $P^{-1}(x)$?

- (A) $P^{-1}(x) = \frac{x+5}{2}$
- (B) $P^{-1}(x) = \frac{x-5}{2}$
- (C) $P^{-1}(x) = \frac{x+5}{3}$
- (D) $P^{-1}(x) = \frac{x-5}{3}$
- (E) $P^{-1}(x) = \frac{x+5}{4}$

Q5. An investor's return on investment (ROI) R (in percent)

is related to the investment amount x (in dollars) by $R(x) = 0.05x + 2$. What is $R^{-1}(x)$?

- (A) $R^{-1}(x) = \frac{x-2}{0.05}$
- (B) $R^{-1}(x) = \frac{x+2}{0.05}$
- (C) $R^{-1}(x) = \frac{x-2}{0.1}$
- (D) $R^{-1}(x) = \frac{x+2}{0.1}$
- (E) $R^{-1}(x) = \frac{x-2}{0.01}$

Q6. Given a tax function $T(x) = 0.2x + 100$, where x is income (in dollars) and $T(x)$ is the tax paid (in dollars), find the inverse function $T^{-1}(x)$.

- (A) $T^{-1}(x) = \frac{x-100}{0.2}$
- (B) $T^{-1}(x) = \frac{x+100}{0.2}$
- (C) $T^{-1}(x) = \frac{x-100}{0.1}$
- (D) $T^{-1}(x) = \frac{x+100}{0.1}$
- (E) $T^{-1}(x) = \frac{x-100}{0.3}$

Q7. What is the stock price S (in dollars) if the number of shares x is related to S by $S(x) = 50 - 2x$? Find $S^{-1}(x)$.

- (A) $S^{-1}(x) = \frac{50-x}{2}$
- (B) $S^{-1}(x) = \frac{50+x}{2}$
- (C) $S^{-1}(x) = \frac{50-x}{3}$
- (D) $S^{-1}(x) = \frac{50+x}{3}$
- (E) $S^{-1}(x) = \frac{50-x}{4}$

Q8. If $f(x) = x^2 + 1$, does f have an inverse that is a function?

- (A) Yes, $f^{-1}(x) = \sqrt{x-1}$
- (B) Yes, $f^{-1}(x) = -\sqrt{x-1}$
- (C) Yes, $f^{-1}(x) = \pm\sqrt{x-1}$
- (D) No, f does not have an inverse that is a function
- (E) None of the above

Open Ended Questions (FRQ)

Q9. Suppose $C(x) = 0.1x + 50$ represents the cost (in dollars) of producing x units of a product. Find $C^{-1}(x)$. Interpret $C^{-1}(x)$ in the context of the problem.

Q10. If the interest rate r (in percent) is related to the initial investment P (in dollars) by $r(P) = 0.03P + 1$, find $r^{-1}(P)$. Use $r^{-1}(P)$ to determine the initial investment if the interest rate is 4

Chef's Tips

- Ensure students have a basic understanding of functions and their inverses.
- Emphasize the importance of checking the inverse by composition.
- Use real-world examples from business and finance to illustrate the practical applications of inverse functions.